MIL-D-10662D
18 July 1983
SUPERSEDING
MIL-D-10662C
4 November 1976

MILITARY SPECIFICATION

1,2-DICHLOROETHANE, TECHNICAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one technical grade of 1,2-dichloroethane, also known as ethylene chloride and ethylene dichloride.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

NN-P-71	-	Pallets, Material Handling, Wood, Stringer Construction, 2-Way and 4-Way (Partial)
PPP-B-585	-	Boxes, Wood, Wirebound
PPP-B-601	-	Boxes, Wood, Cleated-Plywood
PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-C-96	-	Cans, Metal, 28 Gage and Lighter
PPP-C-2020	-	Chemicals, Liquid, Dry, and Paste: Packaging Of
PPP-D-729	-	Drums, Shipping and Storage, Steel, 55-Gallon (208 Liters)
PPP-F-320	-	Fiberboard: Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-TSC-S, Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6810

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

CODE OF FEDERAL REGULATIONS (CFR)

49 CFR 171 to 179 - Department of Transporation Hazardous Materials Regulations

(The Code of Federal Regulations is available from the Superintendent of Documents, US Government Printing Office, Washington, DC 20402. Orders for the above publications should cite "49 CFR 171 to 179.")

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

ASTM STANDARDS

D86 - Distillation of Petroleum Products
D941 - Density and Relative Density (Specific Gravity) of Liquids by Lipkin Bicapillary Pycnometer
D1193 - Reagent Water

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

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3. REQUIREMENTS

3.1 Appearance. 1,2-Dichloroethane shall be free from suspended matter and sediment when tested as specified in 4.2.4.1.

3.2 Chemical and physical characteristics. 1,2-Dichloroethane shall conform to the chemical and physical characteristics of table I when tested as specified therein.

TABLE I. Chemical and physical characteristics

Characteristic	Requirement	Test paragraph
Color	To pass test	4.2.4.2
Boiling range: percent by volume distilling between 82.5° and 84.5°C at 760 mm, minimum	95	4.2.4.3
Specific gravity at 20°/20°C	1.253 to 1.257	4.2.4.4
Acidity (as HCl), percent by weight, maximum	0.005	4.2.4.5

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the 1,2-dichloroethane produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.2).

4.2.2 Sampling.

4.2.2.1 For examination of packaging. Sampling shall be conducted in accordance with MIL-STD-105.

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4.2.2.2 For 1,2-dichloroethane test. See 6.4 for sampling and testing precautions. Sampling shall be conducted in accordance with table II. A representative specimen of approximately 1 liter (L) shall be removed from each sample container and placed in a suitable clean, dry container labeled to identify the lot and container from which it was taken.

TABLE II. Sampling for 1,2-dichloroethane

Number of containers in batch or lot	Number of sample containers
3 to 150	3
151 to 1,200	5
1,201 to 7,000	8
7,001 to 20,000	10
Over 20,000	20

4.2.2.3 For container leakage test. Sampling shall be conducted in accordance with MIL-STD-105, level S-2.

4.2.3 Inspection procedure.

4.2.3.1 For examination of packaging. The sample unit shall be one filled unit or shipping container, as applicable, ready for shipment. Sample unit and shipping containers shall be examined for the following defects using an AQL of 4.0 percent defective:

(a) Contents per container not as specified
(b) Container not as specified
(c) Container closure not as specified
(d) Container damaged or leaking
(e) Fiberboard liners, pads or partitions missing or not as specified
(f) Unitization not as specified
(g) Marking incorrect, missing, or illegible

4.2.3.2 For 1,2-dichloroethane test. Each sample specimen taken in 4.2.2.2 shall be tested as specified in 4.2.4. Failure of any test by any specimen shall be cause for rejection of the lot represented.

4.2.3.3 For container leakage test. The sample unit shall be one container. The sample containers selected in 4.2.2.3 shall be tested as specified in 4.2.5 using an AQL of 1.5 percent defective.

4.2.4 1,2-Dichloroethane tests. See 6.6 for sampling and testing precautions. Water in accordance with ASTM D1193 and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

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4.2.4.1 Appearance. Visually examine the specimen for suspended matter.

4.2.4.2 Color. Prepare a color standard by dissolving 0.0030 gram of potassium dichromate in 1 L of water. Fill two matched Nessler tubes to a height of approximately 24 centimeters with the specimen and the color standard, respectively. Cover the sides of the tubes with opaque paper to eliminate side light. Place the tubes side by side on a white background and view them from a point directly above. The specimen shall be no darker than the color standard.

4.2.4.3 Boiling range. Determine the boiling range of the specimen in accordance with ASTM D86.

4.2.4.4 Specific gravity. Determine the specific gravity of the specimen at 20°/20°C in accordance with ASTM D941.

4.2.4.5 Acidity. Transfer 25 milliliters (mL) of the specimen into a 250-mL glass-stoppered Erlenmeyer flask and add 25 mL of water. Place the stopper in the flask and shake for 5 minutes. Titrate with 0.01N sodium hydroxide solution. Calculate the percent by weight acidity as hydrochloric acid as follows:

$$\text{Percent acidity} = \frac{3.646\ AB}{CD}$$

where: A = Milliliters of sodium hydroxide solution used in the titration,
B = Normality of the sodium hydroxide solution used,
C = Milliliters of specimen used, and
D = Specific gravity of the specimen determined in 4.2.4.4.

4.2.5 Container leakage test. Subject the containers to the rough handling test specified in PPP-C-2020.

4.2.5.1 Drum leakage. Roll the drum in four 90° increments, holding each position for 5 minutes. Observe the sides, bottoms, and tops for any evidence of leakage.

5. PACKAGING (See 6.5)

5.1 Unit packing. Dichloroethane shall be unit packed level A or C in accordance with Department of Transportation (DOT) regulations and as specified below (see 6.2).

5.1.1 Level A. Dichloroethane shall be unit packed level A in a 1- or 55-gallon (gal) quantity as specified (see 6.2).

5.1.1.1 One-gal quantity. A quantity of 10.4 (+0.1 or -0) pounds (lb) of dichloroethane shall be unit packed in accordance with the type II, class 1

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requirements of PPP-C-2020 in a nominal 1-gal capacity rectangular metal can conforming to type V, class 4 of PPP-C-96. The can shall be formed from tinplate, minimum number 100, and shall be coated plan B, with side seam striked. Screw-cap liner, facing, and can-seam sealant shall be resistant to dichloroethane. The screw-cap of the can shall be torqued closed to within a range in accordance with the can supplier's recommendations. A tamper-deterrent inner seal shall be furnished and an anti-back off device shall be applied to the screw-cap. There shall be no evidence of leakage of the can when tested as specified in 4.2.5.

5.1.1.2 Fifty-five-gal quantity. A quantity of 572 (+5 or -0) lb of dichloroethane shall be unit packed in a nominal 55-gal capacity steel drum conforming to type I, class A of PPP-D-729. The drum shall be clean, free of corrosion, and dry. The gaskets for the closures and the sealing compound for the seams of the drum shall be resistant to dichloroethane. The threaded plugs shall be closed to a torque within the range specified by the drum manufacturer. There shall be no leakage of contents from the drum when tested as specified in 4.2.5. A pilfer deterrent cap-seal, color optional, shall be secured over each closure after the leak testing procedure is completed.

5.1.2 Level C. A quantity of 10.4 (+0.1 or -0) lb or 572 (+5 or -0) lb of dichloroethane as specified (see 6.2), shall be unit packed in accordance with the applicable commercial requirements of PPP-C-2020.

5.2 Packing. Dichloroethane shall be packed level A, B or C in accordance with DOT regulations and as specified below (see 6.2).

5.2.1 Level A.

5.2.1.1 One-gal quantity. Four 1-gal cans of dichloroethane unit packed as specified in 5.1.1.1 shall be packed closure end uppermost in one layer in a close-fitting wood box. The box shall conform to class 2, style 4, grade A, overseas type of PPP-B-621, overseas type; style A, B, I or J, grade A of PPP-B-601; or class 3, style optional of PPP-B-585 with preservative treatment as specified for the grade A box of PPP-B-621. Each box shall be for a type 2 average load. All inside surfaces of the box shall be lined with fiberboard and each can shall be inserted in a close-fitting full-can-height cell formed from fiberboard half-slotted partitions. Fiberboard pads shall be inserted where needed to prevent motion of contents. The box shall be closed and reinforced as specified in the applicable box specification. Fiberboard liners, pads and partitions shall be formed from fiberboard conforming to grade W5c or V3c of PPP-F-320.

5.2.1.2 Fifty-five-gal quantity. The 55-gal quantity, unit packed as specified in 5.1.1.2, shall require no further protection for shipment.

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5.2.2 Level B.

5.2.2.1 One-gal quantity. Four 1-gal cans of dichloroethane shall be packed level B in the same manner as is specified for level A protection in 5.2.1.1, except that the box shall be fiberboard conforming to grade V3c of PPP-B-636.

5.2.3 Level C. The 1- or 55- gal quantity of dichloroethane, unit packed as specified in 5.1, shall be packed level C in accordance with the applicable commercial requirements of PPP-C-2020.

5.3 Unitization. Level A and B packs of 1-gal cans of dichloroethane shall be palletized in accordance with load type I of MIL-STD-147, using the softwood pallet conforming to type IV, grade A of NN-P-71. Level C packs shall be unitized in a manner to assure safe delivery from supplier to first destination, stablility in stacking during storage, and acceptance by the carrier. Unitized loads shall be in conformance with Uniform Freight Classification Rules, National Motor Freight Classification Rules, or the carrier rules for other intended modes of transporation.

5.4 Marking. Level A, B and C containers and unitized loads shall be marked in accordance with MIL-STD-129 and DOT regulations. Each container shall be marked to show the lot number and date of manufacture of the dichloroethane, and the following precautionary marking:

HAZARDS:
CANCER SUSPECT AGENT
FLAMMABLE LIQUID
Toxic by Inhalation
Toxic by Absorption

WARNING! CANCER SUSPECT AGENT, FLAMMABLE, HARMFUL IF INHALED OR ABSORBED.

Keep away from heat, sparks, and open flame.
Keep container closed.
Use only with adequate ventilation.
Poisonous gases are produced in fire.
Extinguish with dry chemical, foam, or carbon dioxide.
Avoid breathing vapor.
Avoid contact with eyes, skin and clothing.
Do not take internally.

FIRST AID: If inhaled, remove to fresh air. If breathing has stopped, give artificial respiration. If breathing is difficult, give oxygen. In case of contact, immediately flush eyes or skin with plenty of water for at least 15 minutes. Remove contaminated clothing and shoes. Call a physician. Wash clothing before reuse. Discard contaminated shoes.

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6. NOTES

6.1 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification
(b) Level of unit packing and packing required (see 5.1 and 5.2)
(c) Unit quantity required (see 5.1.1 and 5.1.2).

6.2 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.3 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value should be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of ASTM E29.

6.4 Sampling and testing precautions. This specification covers inspection of chemical material (1,2-dichloroethane, cancer suspect agent, flammable liquid, toxic by inhalation, toxic by absorption) which is potentially hazardous to personnel. All applicable safety rules, regulations, and procedures must be followed in the handling and processing of this material.

6.5 Regulatory coverage. The packaging requirements specified herein are based on regulations current at the time of specification preparation. Because regulations may change, it is recommended that the applicable regulations be reviewed at the time of 1,2-dichloroethane packaging.

Custodians:

Army - EA
Air Force - 68

Review activities:

Army - MI, MD
DLA - GS

Preparing activity:

Army - EA

Project No. 6810-B391

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions -- Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-D-10662D	1,2-DICHLOROETHANE, TECHNICAL

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation.

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* -- Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* -- Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* -- Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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